

Blood-Brain Barrier Gene Conservation Across Human, Mouse, and Macaque

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Introduction

The standard preclinical pipeline for drugs targeting the brain looks like this:

Mouse → validate → **Macaque** → validate → **Human clinical trial**

Each arrow is a translational leap. A drug that works in mouse must survive the jump to macaque, then to human. If the underlying molecular machinery — the blood-brain barrier (BBB) genes that control what enters and exits the brain — differs substantially between species, that pipeline is fragile. If it is conserved, the pipeline has a stronger molecular foundation.

This project asks:

How similar are BBB genes across all three rungs of this translational ladder — at the DNA sequence level?

We quantify this using two complementary metrics:

- **% nucleotide identity** — how many DNA letters match between species
- **dN/dS ratio** — whether evolution is actively *protecting* these genes from protein-changing mutations (dN/dS $\ll$ 1 = purifying selection)

The analysis covers three pairwise comparisons: Human vs Mouse, Human vs Macaque, and Mouse vs Macaque. BBB genes are compared against a liver control set to establish whether any conservation signal is BBB-specific or just a property of all expressed genes.

What are “canonical BBB genes”?

The BBB is formed by brain endothelial cells (ECs) — the cells lining the brain’s blood vessels. “Canonical BBB genes” refers to genes that are *consistently identified across multiple independent transcriptomic studies* as highly expressed and functionally important in these cells. This includes:

- **Tight junction proteins** — CLDN5, OCLN, TJP1 (seal gaps between cells to block free diffusion into the brain)
- **Transporters** — SLC2A1/GLUT1 (glucose), ABCB1/P-gp (drug efflux pump), SLC7A5 (amino acids)
- **Signalling receptors** specific to brain endothelium

“Canonical” does not mean “the only important BBB genes” — it means well-validated across datasets.

Pipeline overview

```

Step 1    Build master BBB gene list (Daneman + Munji)
Step 2    Process Wälchli single-cell EC expression
Step 3    Map gene symbols → Ensembl IDs [HomoloGene attempt - superseded]
Step 3b   Re-run mapping via BioMart + add orthology metadata
Step 4    Download GTF genome annotations [removed - unnecessary]
Step 4b   Cross-reference against human EC datasets
Step 4c   Build liver control set from Human Protein Atlas
Step 5a   Download CDS (coding sequence) FASTAs
Step 5b   Extract one CDS per gene (longest transcript)
Step 5b*  Alternative: Ensembl Canonical / MANE Select transcripts
Step 5c   Pairwise alignment + % identity + dN/dS for BBB genes
Step 5d   Same for liver control genes
Step 5e   Statistical comparison (Wilcoxon)
Step 5f   Recompute dN/dS with codon-aware alignment [bug fix]
Step 5g   Multiple testing correction
Step 5h   Direct mouse vs macaque comparison

```

Note on step numbering: Steps jump from 3 to 3b and from 4 to 4b — these reflect the revision history. See the relevant tabs for details.

Prerequisites

Read-only vs reproducible

This HTML document is **read-only** — you can read the code and results but cannot run it here. To reproduce the analysis from scratch, open the scripts in RStudio (or VS Code with the R extension) and run each **scripts/step*.R** file in order.

The heavy computation (genome alignments, BioMart queries) was already run; this tutorial reads those saved outputs.

```
suppressPackageStartupMessages({  
  library(readr)  
  library(dplyr)  
  library(ggplot2)  
  library(knitr)  
})  
knitr::opts_knit$set(root.dir = "..")
```

R packages

Install all required packages before running any scripts:

```
# CRAN packages  
install.packages(c(  
  "readr", "dplyr", "ggplot2", "tidyr", "knitr",  
  "readxl", "homologene", "seqinr"  
))  
  
# Bioconductor packages  
if (!requireNamespace("BiocManager", quietly = TRUE))  
  install.packages("BiocManager")  
  
BiocManager::install(c(  
  "biomaRt",      # Ensembl cross-species orthologue queries  
  "Biostrings",  # reading/writing FASTA files, DNA translation
```

```

"pwalign"      # pairwise sequence alignment (Needleman-Wunsch)
))

# Seurat - required for Step 2 only
install.packages("Seurat")

```

Raw data files

File	Where to get it	Place it at
Daneman 2010 S3	PLOS ONE paper → Supplementary S3	raw_data/daneman_2010/Daneman2010_S3-Co
Daneman 2010 S4	Same paper → Supplementary S4	raw_data/daneman_2010/Daneman2010_S4-Br
Daneman 2010 S5	Same paper → Supplementary S5	raw_data/daneman_2010/Daneman2010_S5-Br
Munji 2019 S5	Nat. Neurosci. paper → Supplementary S5	raw_data/munji_2019/Munji2019_S5_BBBEnr
Wälchli 2024 Seurat object	Nature paper → repository	raw_data/walchli_2024/Walchli2024_Adult
HPA RNA consensus	Auto-downloaded by step4c	raw_data/hpa_liver/rna_tissue_consensus
Ensembl CDS FASTAs (×3)	Auto-downloaded by step5a	genomes/Homo_sapiens.GRCh38.113.cds.all etc.

Directory structure

```

BBB-Gene-Conservation/
  raw_data/
    daneman_2010/      ← Daneman .xls files
    munji_2019/        ← Munji .xlsx file
    walchli_2024/      ← Seurat .rds.gz object
    hpa_liver/         ← HPA .tsv (auto-downloaded)
  genomes/            ← CDS FASTAs (auto-downloaded; not tracked by git)
  processed/          ← All pipeline outputs (CSVs)
  scripts/            ← All computation scripts

```

Step 1 — Gene List

Why did we do Step 1?

Before we can measure anything, we need to know *which genes* to measure. The BBB does not have one official gene list — different research groups have profiled brain endothelial cells at different times using different technologies. Step 1 builds a single, well-annotated master list from the two most-cited published datasets, tags each gene by how stringently it was identified, and converts mouse gene symbols to human ones so the whole pipeline can work in human gene space.

Data flow

```
Full script: scripts/step1_extract_BBB_genelist.R
Reads: raw_data/daneman_2010/Daneman2010_S3/S4/S5_*.xls,
raw_data/munji_2019/Munji2019_S5_*.xlsx Writes: processed/master_BBB_genelist.csv
Packages: readxl, dplyr, readr, homologene
```

Why Daneman + Munji

Two foundational papers each profiled brain endothelial cells in mouse:

- **Daneman et al. 2010** — compared brain ECs to peripheral, liver, and lung ECs (supplementary tables S3, S4, S5). The S3 list is the strictest (genes enriched vs both liver AND lung).
- **Munji et al. 2019** — top CNS endothelial-enriched genes (supplementary table S5).

Mouse gene symbols from both papers were converted to human HGNC symbols using the `homologene` R package. Each gene was tagged by source paper and which Daneman filter it passed (`liver_and_lung`, `liver_only`, or `lung_only`).

Why we expanded to S3+S4+S5

Initial work used only the strict S3 list (~545 genes). Canonical BBB genes like **SLC2A1** (GLUT1, the main brain glucose transporter) were missing — S3 demands enrichment vs liver AND lung simultaneously, and SLC2A1 is also expressed in liver endothelium. Including S4 and S5 with stringency tags captured these missing genes while preserving the ability to filter back to S3 for sensitivity analysis.

Result

```
master <- read_csv("processed/master_BBB_genelist.csv", show_col_types = FALSE)

cat("Total genes in master list:", nrow(master), "\n\n")
```

Total genes in master list: 1526

```
table(master$Source) |>
  as.data.frame() |>
  rename(Source = Var1, Genes = Freq) |>
  kable()
```

Source	Genes
Both	80
Daneman	1109
Munji	337

```
master |>
  filter(!is.na(Daneman_Filter)) |>
  count(Daneman_Filter) |>
  kable(caption = "Daneman stringency breakdown - `liver_and_lung` is the strictest")
```

Table 1: Daneman stringency breakdown <U+2014> liver_and_lung is the strictest

Daneman_Filter	n
liver_and_lung	130
liver_only	492
lung_only	567

Step 2 — Wälchli

Why did we do Step 2?

The BBB gene list from Step 1 was built entirely from **mouse** data. Before trusting it as a human BBB gene list, we need to check: are these genes actually expressed in human brain

endothelial cells? The Wälchli 2024 dataset is the largest and most recent single-cell atlas of the human brain vasculature, making it the best available source for that validation. Step 2 collapses the 30,000+ individual cells into per-cell-type average expression values — a format we can easily query for any gene.

Data flow

Full script: `scripts/step2_process_walchli_seurat.R` **Reads:** `raw_data/walchli_2024/Walchli2024_..._SeuratObject.rds.gz` **Writes:** `processed/Walchli2024_AverageExpression.csv` **Packages:** Seurat, dplyr, readr
Note: Loading the Seurat object requires ~8 GB RAM and several minutes.

The Wälchli 2024 dataset is a Seurat object containing 30,000+ human brain endothelial cells annotated by EC subtype (capillary, arteriole, venule, etc.).

What we did

Used Seurat's `AverageExpression()` to collapse all cells into per-cell-type mean expression values, producing a gene $\times$ cell-type matrix.

The trickiest part: Seurat's default cell identities in this object were patient ages (15, 19 years), not cell types. Running `AverageExpression()` without overriding this would have averaged across patients, not cell types. Fix: `group.by = "ECclusters"`.

Result

```
walchli <- read_csv("processed/Walchli2024_AverageExpression.csv",
                  show_col_types = FALSE)

cat("Wälchli expression matrix:", nrow(walchli), "genes x",
    ncol(walchli) - 1, "cell types\n\n")
```

Wälchli expression matrix: 26666 genes x 12 cell types

```
cat("Cell types:\n")
```

Cell types:


```
cat(paste("  -", colnames(walchli)[-1], collapse = "\n"))
```

- Angiogenic.capillary
- Arteriole
- Artery
- Capillary
- EndoMT
- Large.artery
- Large.vein
- Mitochondrial
- Proliferating.cell
- Stem.to.EC
- Vein
- Venule

The **Mitochondrial** cluster was excluded from downstream analysis — cells with abnormally high mitochondrial gene content are typically dying or low-quality.

Steps 3 & 3b — Mapping

Why did we do Steps 3 & 3b?

To compare DNA sequences across species, we need a stable, species-specific identifier for each gene — not the gene symbol (which changes between species, e.g. human *SLC2A1* vs mouse *Slc2a1*), but a permanent database ID. Ensembl gene IDs (ENSG... for human, ENSMUSG... for mouse, ENSMUG... for macaque) serve this purpose. This step also retrieves orthologue relationships: which mouse gene corresponds to which human gene, and whether it is a clean one-to-one match or a more complex one-to-many (gene duplication) case. That distinction matters a lot for interpreting conservation scores.

Data flow

Full scripts: scripts/step3_map_ensembl_crossspecies.R (super-seded), scripts/step3b_biomart_orthology_confidence.R **Reads:** processed/master_BBB_genelist.csv → processed/master_BBB_genelist_validated.csv + processed/BBB_genes_crossspecies_ensembl.csv **Writes:** processed/master_BBB_genelist_complete.csv **Packages:** biomaRt, dplyr, readr
Note: BioMart queries require internet and take 5–15 minutes. Results are cached.

⚠ What went wrong — and why there is no Step 3a

The first attempt used the **homologene** R package — an offline NCBI orthologue database — to map all three species simultaneously. This worked for human → mouse but **returned no results for macaque**: HomoloGene does not include rhesus macaque (*Macaca mulatta*).

Step 3b re-ran everything via **Ensembl BioMart**, which supports all three species and additionally returns orthologue type and pre-computed % sequence identity. This is why the pipeline skips from Step 3 to Step 3b.

What we did

Three queries to Ensembl BioMart (each uses a different attribute page, so they cannot be combined):

1. Human gene symbol → human Ensembl gene ID (ENSG...)
2. Human Ensembl ID → mouse orthologue (ENSMUSG...)
3. Human Ensembl ID → macaque orthologue (ENSMMUG...)

Step 3b added `orthology_type` and `perc_id` per orthologue.

Result

```
combined <- read_csv("processed/master_BBB_genelist_complete.csv",
                     show_col_types = FALSE)

cat("Total rows in master table:", nrow(combined), "\n")
```

Total rows in master table: 1724

```
cat("  Human Ensembl IDs:  ", sum(!is.na(combined$HumanEnsembl)), "\n")
```

Human Ensembl IDs: 1632

```
cat("  Mouse orthologues:  ", sum(!is.na(combined$MouseEnsembl)), "\n")
```

Mouse orthologues: 1541

```
cat(" Macaque orthologues: ", sum(!is.na(combined$MacaqueEnsembl)), "\n")
```

Macaque orthologues: 1474

```
combined |>
  filter(!is.na(Mouse_OrthoType)) |>
  count(Mouse_OrthoType) |>
  kable(caption = "Mouse orthologue type distribution")
```

Table 2: Mouse orthologue type distribution

Mouse_OrthoType	n
ortholog_many2many	18
ortholog_one2many	196
ortholog_one2one	1327

The 1,724 rows (vs 1,526 input genes) reflects **one-to-many** orthologues — e.g. human ABCB1 maps to both mouse *Abcb1a* and *Abcb1b*. Both rows are kept and flagged.

Step 3c — Biotype audit

Data flow

Full script: scripts/step3c_biotype_check.R
 genomes/*.cds.all.fa.gz (×3 species, already down-loaded), processed/master_BBB_genelist_complete.csv
Reads: processed/gene_biotype_check.csv, processed/gene_biotype_summary.txt
Writes: processed/gene_biotype_summary.txt
Packages: Biostrings, dplyr, readr **Note:** Biotypes are parsed directly from the gene_biotype: field in Ensembl FASTA headers — no BioMart query needed.

Why did we do Step 3c?

We need to confirm that every gene that entered the CDS alignment pipeline is actually **protein-coding**. If any noncoding genes (lncRNAs, pseudogenes, T-cell receptor segments, etc.) were in the master list and somehow extracted a sequence, the analysis would not correctly reflect conservation of BBB coding sequences. This step audits every Ensembl ID in the master list against the **gene_biotype** annotation embedded in the FASTA headers and checks whether any flagged genes appear in the BBB CDS files.

Result

```
biotype_check <- read_csv("processed/gene_biotype_check.csv",
                          show_col_types = FALSE)

# Biotype breakdown for human
bt_human <- biotype_check |>
  filter(!is.na(Human_Biotype)) |>
  count(Human_Biotype, name = "Genes") |>
  arrange(desc(Genes))

kable(bt_human, caption = "Human gene biotype distribution in BBB master list")
```

Table 3: Human gene biotype distribution in BBB master list

Human_Biotype	Genes
protein_coding	1631

```
n_noncoding <- sum(biotype_check$Any_NonCoding, na.rm = TRUE)
n_total      <- nrow(biotype_check)

cat(sprintf("Genes with any noncoding annotation: %d / %d\n", n_noncoding, n_total))
```

Genes with any noncoding annotation: 0 / 1724

```
if (n_noncoding == 0) {
  cat("\nAll mapped genes are protein_coding in all three species.\n")
  cat("Pipeline integrity: CONFIRMED - no noncoding genes entered the CDS alignment.\n")
} else {
  biotype_check |>
    filter(Any_NonCoding) |>
    select(GeneSymbol_Human, HumanEnsembl, Human_Biotype,
           Mouse_Biotype, Macaque_Biotype) |>
    kable(caption = "Noncoding genes flagged in BBB master list")
}
```

All mapped genes are protein_coding in all three species.
Pipeline integrity: CONFIRMED <U+2014> no noncoding genes entered the CDS alignment.

Audit outcome

Every gene successfully mapped to an Ensembl ID is annotated as **protein_coding** in all three species. Genes with no Ensembl ID (93 human, 185 mouse, 250 macaque) were already excluded from the alignment steps because they had no CDS entries in the Ensembl FASTA files. **No noncoding genes entered the conservation scoring pipeline.**

Steps 4b & 4c — Validation

Why did we do Steps 4b & 4c?

Two separate questions are answered here:

Step 4b — The gene list comes from mouse studies, but we are ultimately interested in the *human* BBB. Before running thousands of alignments, we want to flag which genes have cross-species expression evidence in human brain endothelium (using three independent human EC datasets). This does not remove any genes — it gives each gene a confidence score (0–3) that can be used for sensitivity analysis later.

Step 4c — The core scientific question is whether BBB genes are *unusually* conserved, not just conserved. To answer that, we need something to compare against. A liver control set provides a biologically meaningful baseline: liver is a highly expressed, well-studied tissue, so if BBB genes are more conserved than liver genes, that difference is real and not just a property of all expressed genes.

What went wrong — and why Step 4 was removed

Step 4 downloaded full **GTF genome annotation files** (~1.5 GB per species, $\times 3 = \sim 4.5$ GB total) with the intention of extracting CDS coordinates manually. After downloading all three we found that Ensembl already provides **pre-extracted CDS FASTA files** (~20 MB per species) — the same information at a fraction of the size. Step 4 was removed entirely, which is why the numbering jumps from Step 3b to Step 4b.

Step 4b — Human EC Validation

Data flow

Full script: scripts/step4b_human_validation.R **Reads:** processed/master_BBB_genelist.csv, processed/Walchli2024_AverageExpression.csv (+ Yang 2022 and Winkler 2022 gene lists, hardcoded) **Writes:** processed/master_BBB_genelist_validated.csv **Packages:** dplyr, readr

The master BBB gene list was built from **mouse** studies. Step 4b checks which genes are also expressed in **human** brain endothelium using three independent datasets:

- **Wälchli 2024** — expressed > 0 in any EC cluster
- **Yang 2022** — 112-gene endothelial subtype marker list
- **Winkler 2022** — EC marker list from a large human brain vascular atlas

Each gene receives a count (0–3) of how many datasets confirm it.

```
combined <- read_csv("processed/master_BBB_genelist_complete.csv",
                      show_col_types = FALSE)

combined |>
  count(Human_BBB_Datasets_N) |>
  rename(`# datasets confirming` = Human_BBB_Datasets_N, Genes = n) |>
  kable(caption = "Human EC validation score distribution")
```

Table 4: Human EC validation score distribution

# datasets confirming	Genes
0	74
1	1320
2	299
3	31

74 genes have zero human EC evidence — they came purely from mouse data and are flagged in downstream analysis.

Step 4c — Liver Control Gene Set

Data flow

Full	script:	scripts/step4c_liver_control_geneset.R	
Reads:		processed/master_BBB_genelist_complete.csv,	
		raw_data/hpa_liver/rna_tissue_consensus.tsv	(auto-downloaded)
		processed/liver_control_genelist.csv	Writes:
	Packages:	dplyr, readr	

Liver was chosen as the control tissue — metabolically active and highly conserved, providing a meaningful non-BBB baseline. Filtered for genes with liver nTPM > 10 that are NOT in the BBB master list.

```
liver_ctl <- read_csv("processed/liver_control_genelist.csv", show_col_types = FALSE)
cat("Liver control set:", nrow(liver_ctl), "genes\n")
```

Liver control set: 7131 genes

```
cat("Liver nTPM - median:", round(median(liver_ctl$Liver_nTPM), 1),
    " mean:", round(mean(liver_ctl$Liver_nTPM), 1), "\n")
```

Liver nTPM <U+2014> median: 31.2 mean: 346.1

Landmark liver genes confirmed present: ALB (198,523 nTPM), CYP3A4, APOB, TTR.

Steps 5a-b — CDS

Why did we do Steps 5a-b?

Conservation analysis works at the DNA sequence level — we need the actual nucleotide letters for each gene in each species. The full genome of a mammal is ~3 billion base pairs, but most of that is non-coding (introns, regulatory regions, intergenic space). We only want the **coding sequence (CDS)** — the stretch of DNA that actually gets translated into protein. Ensembl provides pre-extracted CDS files for every annotated transcript, which are ~150× smaller than the full genome files and much faster to work with. Step 5b then extracts just the CDS sequences for our specific BBB and liver gene lists, ready for alignment.

i Data flow

Full scripts: scripts/step5a_download_cds_fastas.R, scripts/step5b_extract_cds_sequences.R **Step 5a** downloads Ensembl CDS FASTAs → genomes/*.cds.all.fa.gz (×3 species) **Step 5b reads:** genome FASTAs + gene lists → **Writes:** genomes/cds_BBB_*.fa and genomes/cds_liver_*.fa (×3 species each) **Packages:** Biostrings, dplyr, readr **Note:** genomes/ is excluded from git (too large). Run Step 5a to populate it before Step 5b.

What we did

Ensembl provides pre-extracted CDS FASTA files for every annotated transcript. For each gene with multiple transcripts (alternative splicing), we kept the **longest CDS** as a heuristic for the full open reading frame.

Alternative: scripts/step5b_alt_canonical_transcripts.R uses **Ensembl Canonical / MANE Select** transcripts instead — one curated transcript per gene, jointly designated by NCBI + EMBL-EBI. The FASTAs are already generated in genomes/cds_canonical_*.fa.

Result

```
library(Biostrings)
bbb_human <- readDNAStringSet("genomes/cds_BBB_human.fa")
cat("Human BBB CDS sequences:", length(bbb_human), "\n")
```

Human BBB CDS sequences: 1433

```
cat("Length: median =", as.integer(median(width(bbb_human))), "bp\n")
```

Length: median = 1449 bp

```
cat("      mean  =", as.integer(mean(width(bbb_human))), "bp\n")
```

mean = 1913 bp

Steps 5c-d — Scoring

Why did we do Steps 5c-d?

This is the core of the project — the step that actually measures conservation. For each BBB gene (and each liver control gene), we align its coding sequence between species and compute two numbers: how similar the DNA letters are (% identity), and whether evolution is actively protecting the gene from protein-changing mutations (dN/dS). Running this for both BBB genes and liver genes separately sets up the statistical comparison in Step 5e: are BBB genes more conserved than a typical expressed gene?

Data flow

Full scripts: scripts/step5c_align_and_score.R (BBB), scripts/step5d_liver_scoring.R (liver) **Reads:** genomes/cds_BBB_*.fa or genomes/cds_liver_*.fa, processed/master_BBB_genelist_complete.csv **Writes:** processed/conservation_scores_BBB.csv, processed/conservation_scores_liver.csv, processed/conservation_scores_combined.csv **Packages:** Biostrings, pwalign, seqinr, dplyr, readr **Note:** Step 5c takes 20–40 min (~1,500 alignments). Step 5d takes 60–90 min (~3,000 alignments).

Pairwise global alignment

For each gene we ran a **Needleman-Wunsch global alignment** between species pairs, then computed % identity and dN/dS:

```
aln <- pwalign::pairwiseAlignment(  
  seq1, seq2, type = "global",  
  substitutionMatrix = nucleotideSubstitutionMatrix(match=1, mismatch=-1),  
  gapOpening = -5, gapExtension = -2  
)  
pct_identity <- pwalign::pid(aln, type = "PID1")  
# PID1 = matches / alignment_length (including gaps)
```

Why global, not local? These are orthologous genes — they should align end-to-end. Local alignment would only find the most similar segment, over-estimating similarity for genes that diverge at the edges.

Why % identity AND dN/dS

- **% identity** — how similar are the letters, simple and interpretable

- **dN/dS** — how often do mutations *change the protein* (non-synonymous, dN) vs leave it unchanged (synonymous, dS). $dN/dS \ll 1$ means evolution is removing protein-changing mutations (purifying selection).

Steps 5e-g — Statistics

Why did we do Steps 5e-g?

Having conservation scores for ~4,500 gene pairs is not enough on its own — we need to know whether the differences between BBB and liver genes are statistically meaningful or just random variation. Step 5e runs the formal tests. Steps 5f and 5g were added specifically because the initial dN/dS values were wrong (see the bug callout below): Step 5f recomputes them correctly using a codon-aware alignment pipeline, and Step 5g re-runs the statistics on the corrected values with proper multiple testing correction applied.

Data flow

Full scripts: `scripts/step5e_statistics.R`, `scripts/step5f_recompute_dnds.R`, `scripts/step5g_stats_v2.R` **Step 5e** reads `conservation_scores_combined.csv` → writes `conservation_summary.txt` **Step 5f** reads combined scores + CDS FASTAs → writes `conservation_scores_combined_v2.csv` **Step 5g** reads v2 scores → writes `conservation_summary_v2.txt` **Packages:** Biostrings, pwalign, seqinr, dplyr, readr

Step 5e: First-pass statistics

Wilcoxon rank-sum tests comparing BBB vs liver (non-parametric, because % identity is bounded and skewed). Stratified analyses by Daneman filter, validation score, and orthologue type.

Step 5f: dN/dS recomputation

What went wrong — three separate dN/dS bugs

Bug 1 — all NA values: The `ape` package (loaded alongside `seqinr`) has its own `kaks()` function and silently overrode `seqinr::kaks()`. Fix: explicit namespace.

Bug 2 — values inflated: `seqinr::kaks()` assumes codon-aligned input — same length, same reading frame positions in both sequences. We were passing unaligned

sequences, inflating the non-synonymous count. Especially large effect for human vs mouse (~80 million year divergence).

Bug 3 — wrong translation function: dplyr's `translate()` (an SQL function) was silently overriding `Biostrings::translate()`. Fix: `Biostrings::translate(DNAString(seq), if.fuzzy.codon = "X")`.

The fix: translate both sequences to amino acids → align proteins with BLOSUM62 → back-map codons using protein alignment → strip gap columns → pass codon-aligned pair to `seqinr::kaks()`:

```
aa1 <- Biostrings::translate(DNAString(nt1), if.fuzzy.codon = "X")
aa2 <- Biostrings::translate(DNAString(nt2), if.fuzzy.codon = "X")
protein_aln <- pwalgn::pairwiseAlignment(aa1, aa2, type = "global",
                                         substitutionMatrix = "BLOSUM62",
                                         gapOpening = 10, gapExtension = 4)
# back-map to codons, strip gaps, run seqinr::kaks()
# full implementation: scripts/step5f_recompute_dnds.R
```

Step 5g: Multiple testing correction

Bonferroni and FDR (Benjamini-Hochberg) applied to the four primary BBB-vs-liver comparisons. Stratified tests are flagged as exploratory.

Final results

```
df <- read_csv("processed/conservation_scores_combined_v2.csv",
               show_col_types = FALSE)

bbb <- df[df$Gene_Set == "BBB", ]
liver <- df[df$Gene_Set == "Liver_Control", ]

results <- data.frame(
  Comparison = c("Human vs Mouse - % identity",
                 "Human vs Macaque - % identity",
                 "Human vs Mouse - dN/dS (corrected)",
                 "Human vs Macaque - dN/dS (corrected)"),
  BBB_median = c(
    round(median(bbb$PctId_Human_Mouse, na.rm = TRUE), 2),
    round(median(bbb$PctId_Human_Macaque, na.rm = TRUE), 2),
```

```

round(median(bbb$dNdS_Human_Mouse_v2,      na.rm = TRUE), 3),
round(median(bbb$dNdS_Human_Macaque_v2,    na.rm = TRUE), 3)
),
Liver_median = c(
  round(median(liver$PctId_Human_Mouse,      na.rm = TRUE), 2),
  round(median(liver$PctId_Human_Macaque,    na.rm = TRUE), 2),
  round(median(liver$dNdS_Human_Mouse_v2,    na.rm = TRUE), 3),
  round(median(liver$dNdS_Human_Macaque_v2,  na.rm = TRUE), 3)
)
)
kable(results, caption = "BBB vs liver conservation - final corrected values")

```

Table 5: BBB vs liver conservation <U+2014> final corrected values

Comparison	BBB_median	Liver_median
Human vs Mouse <U+2014> % identity	84.120	84.320
Human vs Macaque <U+2014> % identity	96.820	96.150
Human vs Mouse <U+2014> dN/dS (corrected)	0.117	0.119
Human vs Macaque <U+2014> dN/dS (corrected)	0.152	0.196

```

df_long <- df |>
select(Gene_Set, PctId_Human_Mouse, PctId_Human_Macaque) |>
tidyr::pivot_longer(cols = -Gene_Set, names_to = "Comparison",
  values_to = "PctIdentity") |>
filter(!is.na(PctIdentity)) |>
mutate(Comparison = sub("PctId_", "", Comparison),
  Comparison = sub("_", " vs ", Comparison))

ggplot(df_long, aes(x = PctIdentity, fill = Gene_Set)) +
  geom_density(alpha = 0.5) +
  facet_wrap(~ Comparison) +
  scale_fill_manual(
    values = c("BBB" = "#d62728", "Liver_Control" = "#1f77b4"),
    labels = c("BBB" = "BBB (red)", "Liver_Control" = "Liver (blue)")
  ) +
  labs(x = "% nucleotide identity", y = "Density",
    title = "BBB vs Liver: distribution of % identity",
    fill = "Gene set",
    caption = "Purple = region where the BBB and Liver distributions overlap") +
  theme_minimal() +
  theme(plot.caption = element_text(colour = "#7b4f6e", face = "italic", size = 9))

```

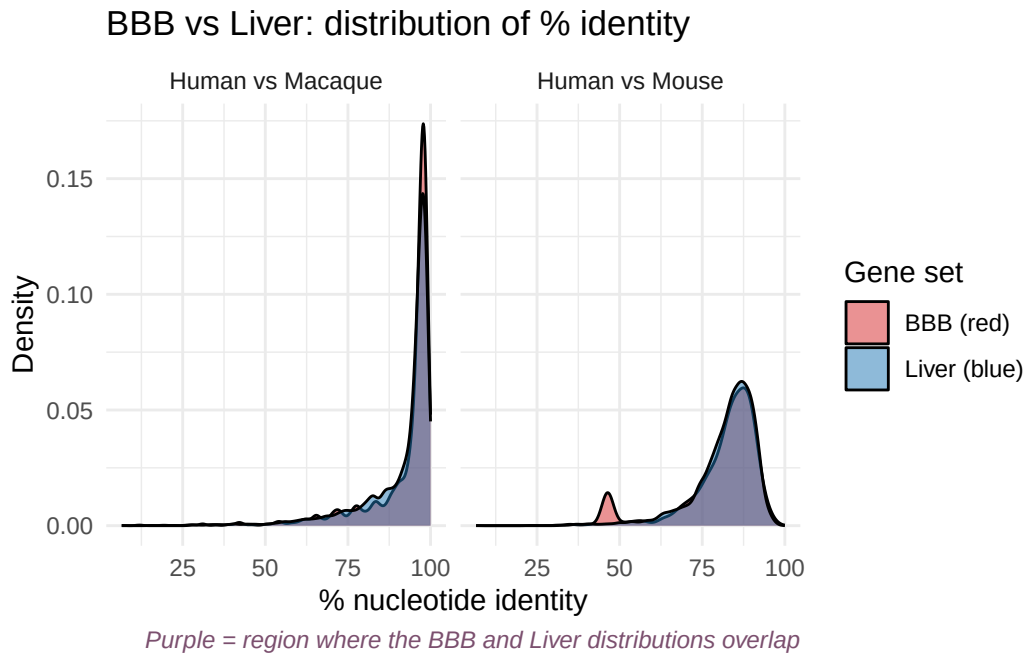


Figure 1: Distribution of % nucleotide identity <U+2014> BBB vs Liver

```
df_dnds <- df |>
  select(Gene_Set, dNdS_Human_Mouse_v2, dNdS_Human_Macaque_v2) |>
  tidyr::pivot_longer(cols = -Gene_Set, names_to = "Comparison", values_to = "dNdS") |>
  filter(!is.na(dNdS), dNdS < 2)

ggplot(df_dnds, aes(x = dNdS, fill = Gene_Set)) +
  geom_density(alpha = 0.5) +
  facet_wrap(~ Comparison) +
  scale_fill_manual(
    values = c("BBB" = "#d62728", "Liver_Control" = "#1f77b4"),
    labels = c("BBB" = "BBB (red)", "Liver_Control" = "Liver (blue)")
  ) +
  labs(x = "dN/dS (codon-aligned)", y = "Density",
       title = "BBB vs Liver: distribution of dN/dS",
       fill = "Gene set",
       caption = "Purple = region where the BBB and Liver distributions overlap") +
  theme_minimal() +
  theme(plot.caption = element_text(colour = "#7b4f6e", face = "italic", size = 9))
```

BBB vs Liver: distribution of dN/dS

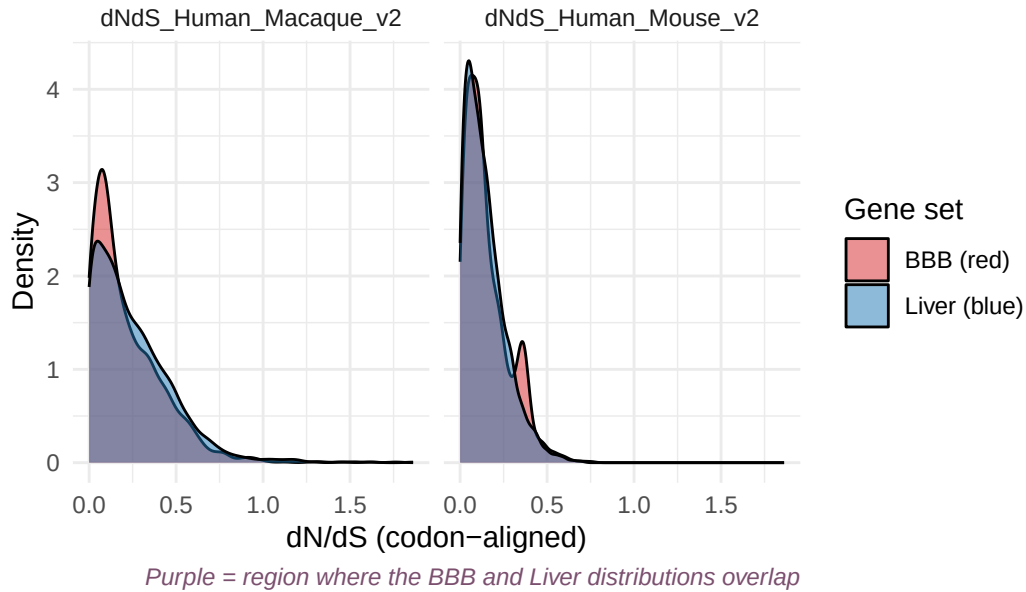


Figure 2: Distribution of dN/dS (codon-aligned)

```
bbb_scatter <- bbb |>
  filter(!is.na(PctId_Human_Mouse), !is.na(dNdS_Human_Mouse_v2), dNdS_Human_Mouse_v2 < 2)

ggplot(bbb_scatter, aes(x = PctId_Human_Mouse, y = dNdS_Human_Mouse_v2)) +
  geom_point(alpha = 0.25, color = "#d62728", size = 1.2) +
  geom_smooth(method = "loess", se = TRUE, color = "black", linewidth = 0.8) +
  labs(x = "% nucleotide identity (Human vs Mouse)", y = "dN/dS (codon-aligned)",
       title = "% identity and dN/dS are correlated but capture different signal",
       subtitle = "BBB genes only - Human vs Mouse") +
  theme_minimal()
```

% identity and dN/dS are correlated but capture different signal
 BBB genes only <U+2014> Human vs Mouse

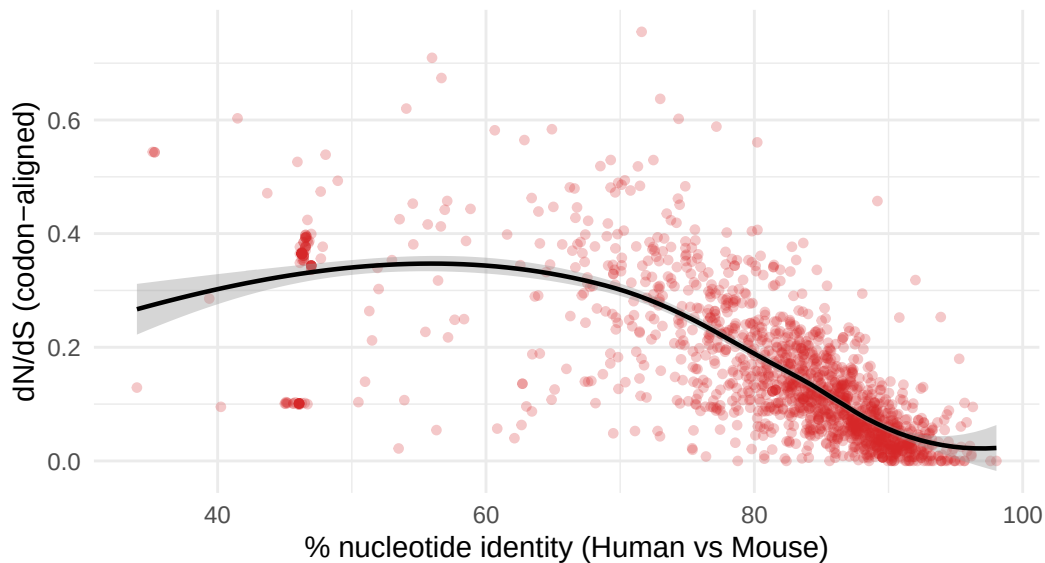


Figure 3: % identity vs dN/dS <U+2014> both metrics tell the same story

```
purify_df <- df |>
  filter(!is.na(dNdS_Human_Mouse_v2) | !is.na(dNdS_Human_Macaque_v2)) |>
  group_by(Gene_Set) |>
  summarise(
    `Human vs Mouse` = mean(dNdS_Human_Mouse_v2 < 0.3, na.rm = TRUE) * 100,
    `Human vs Macaque` = mean(dNdS_Human_Macaque_v2 < 0.3, na.rm = TRUE) * 100,
    .groups = "drop"
  ) |>
  tidyr::pivot_longer(cols = -Gene_Set, names_to = "Comparison", values_to = "Pct")

ggplot(purify_df, aes(x = Comparison, y = Pct, fill = Gene_Set)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c("BBB" = "#d62728", "Liver_Control" = "#1f77b4")) +
  labs(x = NULL, y = "% of genes with dN/dS < 0.3",
       title = "Proportion of genes under strong purifying selection",
       subtitle = "dN/dS < 0.3 = conventional threshold for strong constraint",
       fill = "Gene set") +
  theme_minimal()
```

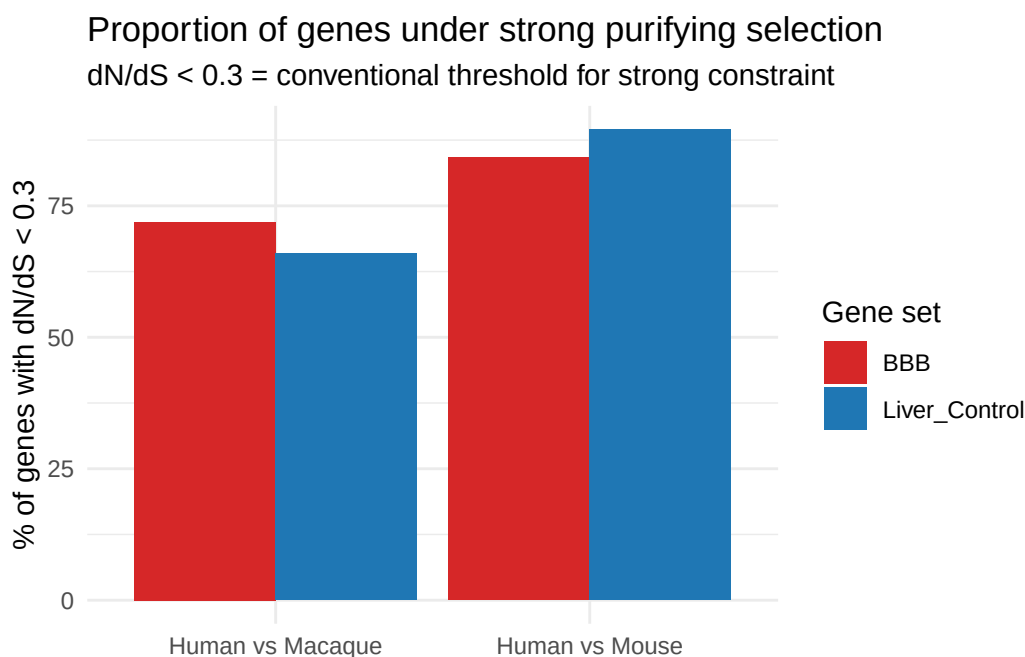


Figure 4: Proportion of genes under strong purifying selection ($dN/dS < 0.3$)

Step 5h — Triangle

Why did we do Step 5h?

Steps 5c–g compared every species against **human** as the reference point. But the translational pipeline runs Mouse → Macaque → Human, which means the Mouse–Macaque leg is just as important as the others. If mouse and macaque are very different from each other, a drug validated in mouse might fail in macaque for reasons unrelated to human biology. Step 5h closes the triangle by computing mouse vs macaque conservation directly, giving us the complete three-way picture needed to evaluate the full translational ladder.

i Data flow

Full	script:	<code>scripts/step5h_mouse_vs_macaque.R</code>	Reads:
<code>genomes/cds_BBB_mouse.fa,</code>	<code>genomes/cds_BBB_macaque.fa,</code>	<code>full</code>	genome
<code>FASTAs,</code>	<code>processed/conservation_scores_combined_v2.csv</code>		Writes:
<code>processed/conservation_scores_mouse_macaque.csv</code>	Note:	<code>~1,500 alignments,</code>	
<code>~30 minutes.</code>			

Earlier steps used human as the reference. To complete the picture we computed the third side: mouse vs macaque directly.

```
mm      <- read_csv("processed/conservation_scores_mouse_macaque.csv",
                    show_col_types = FALSE)
mm_bbb <- mm[mm$Gene_Set == "BBB", ]

triangle <- data.frame(
  Comparison = c("Human vs Mouse", "Human vs Macaque", "Mouse vs Macaque"),
  `Median % identity (BBB genes)` = c(
    round(median(bbb$PctId_Human_Mouse,      na.rm = TRUE), 1),
    round(median(bbb$PctId_Human_Macaque,     na.rm = TRUE), 1),
    round(median(mm_bbb$PctId_Mouse_Macaque, na.rm = TRUE), 1)
  ),
  check.names = FALSE
)
kable(triangle)
```

Comparison	Median % identity (BBB genes)
Human vs Mouse	84.1
Human vs Macaque	96.8
Mouse vs Macaque	82.6

```
bbb_both <- bbb |>
  filter(!is.na(PctId_Human_Mouse), !is.na(PctId_Human_Macaque))

ggplot(bbb_both, aes(x = PctId_Human_Mouse, y = PctId_Human_Macaque)) +
  geom_point(alpha = 0.25, color = "#d62728", size = 1.2) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed",
             color = "grey40", linewidth = 0.8) +
  annotate("text", x = 60, y = 98, label = "Macaque closer to human",
          color = "grey30", size = 3.5, hjust = 0) +
  annotate("text", x = 85, y = 72, label = "Mouse closer to human",
          color = "grey30", size = 3.5, hjust = 0) +
  labs(x = "Human vs Mouse % identity", y = "Human vs Macaque % identity",
       title = "For almost every BBB gene, macaque is more similar to human than mouse is",
       subtitle = "Points above the dashed diagonal = macaque is the closer model") +
  theme_minimal()
```

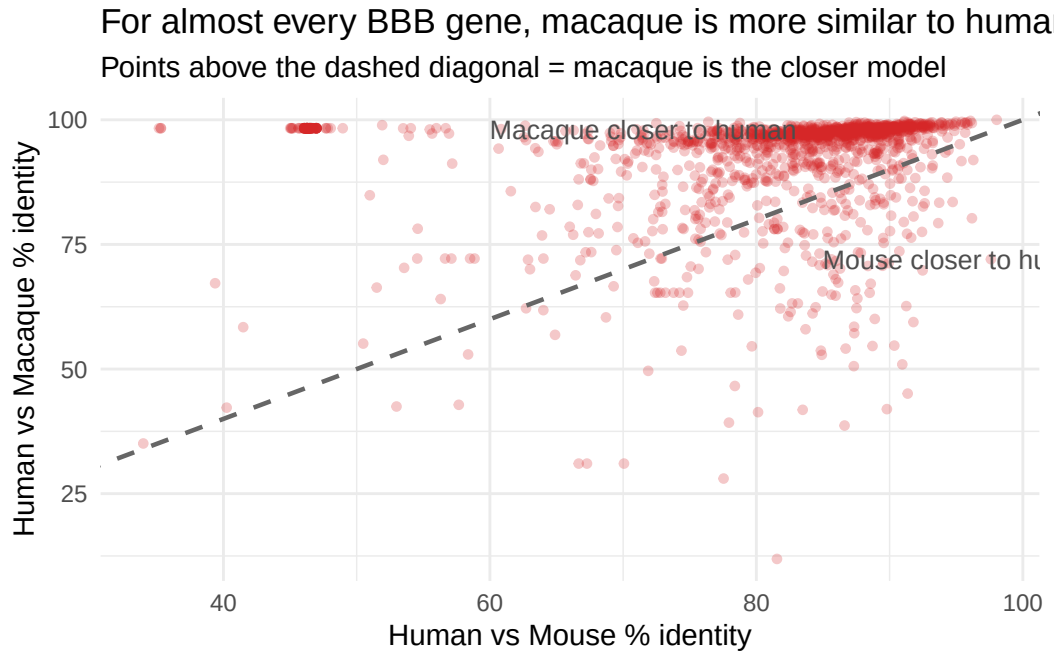
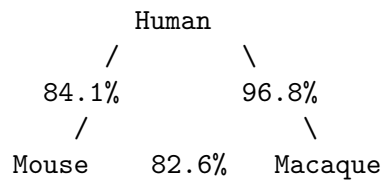


Figure 5: Human vs Mouse and Human vs Macaque % identity per BBB gene

The complete evolutionary distance picture:



Results

Headline findings

1. **Macaque is the closest model to human BBB.** 96.8% nucleotide identity at the coding-sequence level. For almost every gene, macaque sits above the diagonal — more similar to human than mouse.

2. **Mouse is more divergent (84.1%) but still highly conserved.** Mouse models remain biologically defensible, but ~16% sequence divergence is present. Where mouse and human BBB phenotypes differ clinically, this divergence is one possible explanation.
3. **BBB genes are under stronger purifying selection than liver genes** — most visibly in the human-macaque comparison (BBB dN/dS 0.152 vs Liver 0.196, $p < 10^{-10}$ after Bonferroni correction). The signal is weaker in human-mouse because at ~80 million years, all genes have accumulated extensive change.
4. **Gene duplication produces dramatic divergence.** Paralogous mouse genes (e.g. *Abcb1a* + *Abcb1b* for human ABCB1) are only ~47% identical to human, vs ~85% for one-to-one orthologues. Effect size is large (rank-biserial $r = 0.72$).
5. **Conservation and human BBB expression are independent.** A gene's validation score (0–3 datasets) does not predict how conserved it is at the sequence level.

Biological interpretation

Strong purifying selection on BBB genes makes biological sense. The BBB controls what enters and exits the brain — broken transporters or tight junctions would let toxins in or block essential nutrients. Evolution actively removes protein-changing mutations from these genes, more aggressively than for liver genes.

The fact that this signal is most visible in the human-macaque comparison — not human-mouse — reflects evolutionary timescale. Subtle differential selection is detectable when species are close enough that not all genes have saturated their substitutions. At ~80 million years for human-mouse, this signal is lost in the noise.

Limitations

Methodological caveats

- **Pairwise dN/dS, not phylogenetic.** A more rigorous analysis uses PAML `codeml` on a multi-species alignment with an explicit phylogenetic tree.
- **Longest-transcript heuristic.** MANE Select / Ensembl Canonical (`step5b_alt_canonical_transcript`) are more standard. The canonical FASTAs are already generated.
- **Multiple testing correction was applied to primary tests only.** Stratified subgroup analyses (by Daneman filter, validation score, paralogue type) are exploratory.

Biological caveats

- **The starting gene list is enrichment-based.** CLDN5 (the canonical BBB tight junction protein) is absent because it failed the differential expression cutoff — it is expressed in all endothelia, not just brain.
 - **Liver is one specific control choice.** A different control (brain non-BBB genes, genes matched by protein domain) could give a different picture.
 - **Coding sequence only.** Regulatory regions (promoters, enhancers, UTRs) — where most cross-species expression-level differences live — were deliberately excluded.
-

References

Datasets

- Daneman R, et al. (2010). The mouse blood-brain barrier transcriptome. *PLOS ONE* 5(10): e13741.
- Munji RN, et al. (2019). Profiling the mouse brain endothelial transcriptome in health and disease. *Nature Neuroscience* 22(11): 1892-1902.
- Wälchli T, et al. (2024). Single-cell atlas of the human brain vasculature. *Nature* 614: 945-954.
- Yang AC, et al. (2022). A human brain vascular atlas reveals diverse mediators of Alzheimer’s risk. *Nature* 603: 885-892.
- Winkler EA, et al. (2022). A single-cell atlas of the normal and malformed human brain vasculature. *Science* 375(6584): eabi7377.

Tools

- Ensembl release 113. <https://www.ensembl.org>
- BioMart (biomaRt R package).
- Human Protein Atlas. <https://www.proteinatlas.org>
- Bioconductor: Biostrings, palign, seqinr.

AI Transparency

This pipeline was developed by Yara Ismail with the assistance of Claude (Anthropic) for R script writing and debugging. All scientific decisions, interpretation, and research design were made by the author under supervision of Dr. Claire Clelland.

Render with `quarto render docs/BBB_Conservation_Tutorial.qmd` from the project root.